

Pretest or not to pretest? A preliminary version of a tool for the virtual character standardization

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ABSTRACT

This work-in-progress paper deals with pretesting of the virtual characters in VR psychological experiments. To make pretesting easier, the tool based on the combination of WebGL technology, Unity engine and psychological questionnaire was created. To test its efficacy we asked 13 judges to rate 15 models differing in terms of realism. The results and future directions in the development of the tool are discussed in this paper.

Index Terms: Virtual Reality—agent—avatar—realism pretesting;

1 INTRODUCTION

The degree to which virtual stimuli are perceived as real by the user is called simulation realism [14]. Realism *per se* is a multi-dimensional construct. One of the approaches [19] distinguishes three types of realism of simulation defined as the degree to which certain aspect of the agent/avatar corresponds to the real human: photographic realism (the appearance of the agent/avatar), behavioral realism (the behavioral aspect) and interactional realism (the interaction/communication aspect). Realism is a fundamental characteristic of the virtual experience and is one of the main factors leading to the effective virtual experience [19]. The research and theory shows its relationship with key characteristics of Virtual Reality experience i.e. presence, copresence and immersion [4, 5, 19]. Achieving them, is usually a goal of both researchers and developers [16, 19, 20] so dealing thoroughly with realism needs to have high importance if we aim to create effective virtual reality experience. This need is reflected in growing interest in realism of VR studies (e.g. [12, 15]).

1.1 The issue of pretesting

As shown above, although realism is one of the fundamental aspects of VR it still does not receive enough attention, especially in the psychological VR experiments. This is expressed in the fact that characteristics of virtual characters (among them realism) are usually not pretested (tests of experimental stimuli before the main study, aiming at standardization and evaluation of the experimental material). Although both pre-testing and testing of the stimuli during the main experiment (i.e. manipulation check) allow for stimuli characteristics evaluation, pre-testing goes further beyond, because it happens prior to the study of interest and therefore enables proper

stimuli selection and its adjustment. As Emmerich and Masuch [7] point out, lack of standardization is one of the difficulties connected to the research in VR conducted using virtual characters (one of the rare exceptions: [10]). The common practice is to state that the particular character had “low” or “high” realism, without any experimental justification for such declaration. In the meta-analysis of social facilitation VR research, conducted by Sterna and associates [17] none of the analysed texts provided any form of detailed information on how the agent/avatar was generated and if it was pretested whatsoever. As the researchers conclude [17] in case of no standardization of the stimuli, each time we discuss virtual characters influence on human behavior we are unsure whether we are comparing the results of studies using similar or different levels of realism. Ultimately, it might lead to the significant decrease in replicability of the studies, which is one of the major threats to the field of social psychology [18]. The notion of pretesting is also embedded in current guidelines for the development of immersive virtual reality software for cognitive neuroscience and neuropsychology [13].

Problem of lack of pretesting might be rooted in the time limit-efficiency trade-off. Researchers might not be eager to pretest the stimuli used in the experiments, because it requires additional work devoted to designing the pilot tests and conducting it. It becomes especially difficult if we deal with state-of-the-art technology such as VR which requires additional development work and previous knowledge of the technology to conduct the experiment *per se*. We think that the current state of affairs might be changed with implementation of simple, easy to use and interactive tools designed to pretest the stimuli used in the experiments. Additional constraint which might prevent researchers from conducting pretests is the COVID-19 *pandemia* which forces the scientists to limit the contact with the participants to necessary minimum. Therefore the tool to be found needs to be web-based (online), enabling conducting pretests remotely.

1.2 Aims

Our work has several aims. Firstly, we want to underscore the important issue of pretesting of the virtual stimuli and especially of the virtual social stimuli i.e. virtual agents. Secondly, we want to encourage the wider audience to perform pretesting in their VR experiments. Thirdly, we want to show our preliminary work on creating a tool which can be used in pretesting.

2 METHODS

2.1 Stimuli

The stimuli consisted of 5 models presenting different levels of behavioral realism (B1-B5), 5 models presenting different levels of photographic realism (F1-F5) and 5 models presenting different levels of interactional realism (I1-I5). As a result, we used 15 models (5 of each type of realism). While creating the models we tried to achieve systematic, small, but influential changes in certain aspects

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of virtual characters. The baseline 3D model was created from human body scan (by Scanlab) and modified for purposes of the project in Character Creator 3 software and Unity engine.

2.1.1 Photographic realism

When it comes to photographic realism we manipulated level of details, numbers of triangles, resolution of textures, render pipelines and shaders. The parameters are compared in tab. 1 (supplementary materials). Thus we obtained 5 models depicted in fig. 2 (supplementary materials). For presenting the virtual person in the highest photographic realism (F5) the following features and assets were used: Unity High Definition Render Pipeline; ShaderLab-generated skin shader from The Heretic: Digital Human asset; hair generated with Hair Studio asset with HDRP-related and VR-specific Hair Studio adjustments, based on a Physically Based Rendering (PBR) model [6]; ScanLab photogrammetric body scan optimized by Realusion for realtime rendering, then modified, adjusted to project needs and exported with Character Creator 3. Lighting conditions were similar for all the models (or corresponding between render pipelines). Differences in brightness are due to different textures and shaders properties. The exact lighting conditions were not possible because even if we applied the light with the same intensity and color temperature, there were different render pipelines, different shaders and in the lowest quality the normal maps were not used.

2.1.2 Behavioral realism

Models presenting different levels of behavioral realism were created by manipulating the presence and smoothness of the virtual human movements as well as adding the eye contact of the F5 version of the agent. Therefore we obtained the following: agent with no movement, agent with stiff and uneven movement, agent with smooth movement, agent with stiff and uneven movement and eye contact, agent with smooth movement and eye contact. They are compared in tab. 1 (supplementary materials). The models were presented in a form of a video-clip (Fig. 1, supplementary materials). For animating the agent we have used animations with Unity Mecanim Humanoid mode. Depending on the needs of the specific levels of realism, these were manually adjusted animations from mixamo.com and few manually adjusted FBX animation conversions from the Carnegie-Mellon University mocap library (<http://mocap.cs.cmu.edu/>) merged with manually calibrated custom Unity editor extension called Realistic Eye Movements.

2.1.3 Interactional realism

When creating models presenting different levels of interactional realism we manipulated the presence of the communication, its content and reference to the participant of the F5 version of the agent. Doing that, resulted in creating: silent agent, agent emitting onomatopoeia, agent saying nonsense sentence, agent saying a meaningful sentence, agent asking participant a meaningful question (tab. 1, supplementary materials). The models were presented in a form of a video-clip as in behavioral realism category. In order to obtain convincing high end visual simulation of the face behavior we have used the manually calibrated iClone 7 Live Face markerless facial motion capture with use of the iPhone 11 Pro TrueDepth Camera system combined with realtime lip-sync based on the realtime speech audio signal from ZOOM H6 recorder serving as a microphone.

2.2 Tool

The main tool for assessing agents' realism was the questionnaire¹. We based the questionnaire items on the way VR Simulation

¹The draft English version of the tool is available at: https://filedn.com/1CW1i7Ldf71YaVnvgK475au/13_11_2020d/index.html. If you want to use the tool in your research, contact us: radoslaw.sterna@gmail.com.

Realism Scale [14] was created and generated our own set of questions specifically dedicated to measuring different aspects of virtual character characteristics.

The questionnaire consists of 4 items for each type of realism measured i.e. photographic, behavioral and interactional realism repeated for each of the models. Additionally after set of 5 models in each type of realism the participants are asked to order the models from the least realistic to the most realistic (see Fig. 2, supplementary materials). Therefore, the tool provides us with two measures: the sequence of models of each type of realism and the realism rating for each model. To see the items of the questionnaire please look at tab. 2 (supplementary materials).

The questionnaire was implemented using in Unity 2019.4.11f1 with WebGL 1.0 Graphics API to ensure widest possible range of device types compatibility. Unity was chosen as a multi-platform industry standard engine for creating 3D and VR content. WebGL was chosen as a platform that enables presenting interactive environments in the browser, although in this early version of the tool only static (photos and videos) material was presented. For making the highest 3D realtime rendering presentation possible on older devices - virtual person snapshots and videos were made with use of high quality 4k body scan with top quality shaders and postprocessing effects under High Definition Render Pipeline in Unity 2020.1.11f1 and also with use of Character Creator 3 adjustments, then after transfer within iClone 7.8 with use of iClone shaders.

The application has fully passed our testing in Google Chrome version 86.0.4240.75 (Official Build) (64-bit), Firefox version 81.0.1 (64-bit), Opera version 71.0.3770.228, Brave Browser version 1.15.72 using Chromium version 86.0.4240.75 (Official Build) (64-bit). At the moment the questionnaire application is not fully compatible with mobile devices and it is not compatible with Safari, Internet Explorer and Edge. The resulting data are collected directly into Google Drive Sheet adjusted to specific needs of the questionnaire.

2.3 Participants and procedure

To test the efficacy of our tool we conducted an experiment on a small sample of participants ($N = 13$; 4F, 9M). The mean age was 26.69 (21-35; $SD = 4.86$). The participants were either psychologists experienced in dealing with virtual environments ($n = 11$) or experienced gamers ($n = 2$). The research received a favourable ethical opinion from the Research Ethics Committee at the Institute of Psychology (Jagiellonian University in Kraków). All participants were asked to complete an online questionnaire regarding the realism of virtual characters. Participants rated each virtual character model and ordered them twice to decrease the level of intra-rater noise. All of the models were presented in a random order in 3 blocks, each dedicated to a different type of realism. Summing up, the sequence of rating the realism blocks was: B-F-I, B-F-I, with randomization inside each block. An approximate time of filling in the form was about 30 minutes.

3 RESULTS

In all of the analyses we followed Harpe's [11] recommendation on analysing Likert-type scales in studies with small sample sizes. Therefore, all of the analyses are conducted with usage of non-parametric tests and medians.

3.1 Reliability

First and foremost we aimed to test whether the created subscales have proper internal consistency. To do that, we calculated Cronbach's alpha for each of the subscales in each of the two measurements. The results show that the subscales' internal consistency are very high for behavioral realism (Cronbach's alpha = .89;.97), photographic realism (Cronbach's alpha = .97;.98) and interactional realism (Cronbach's alpha = .94;.96). Furthermore, we wanted to test the test-retest reliability of the scale by correlating the scores

from the first rating round and the second. For behavioral ($r_s = .86$, $p < .001$, $N = 65$), photographic ($r_s = .89$, $p < .001$, $N = 65$) and interactional ($r_s = .76$, $p < .001$, $N = 65$) realism subscales the correlation between two rounds was high.

3.2 Comparison of the models

To test statistically the differences between the models we decided to calculate the median of the two measurements for each item, then calculate the median from 4 items of each type of realism for each virtual character model and each participant and after that use the Friedman test of differences among repeated measures with data split for each type of realism, agent model as a factor and median realism rating as an outcome measure. The analyses revealed main effect of agent model on the rating. The results were highly significant for all of the realism categories: behavioral ($Chi^2 = 17.10$, $p < .001$), photographic ($Chi^2 = 76.18$, $p < .001$) and interactional ($Chi^2 = 22.77$, $p < .001$).

3.3 Ordering

When it comes to the ordering within each type of realism we decided to calculate the frequency of each type of ordering summing both first and second ordering of each type of realism. Therefore overall number of orderings was 6 for each of the participants.

The most common sequences when it comes to the realism of behavior were $B2 < B1 < B3 < B4 < B5$ and $B1 < B2 < B3 < B4 < B5$ with equal number of 5 votes each. In photographic realism the most popular were $F1 < F2 < F3 < F4 < F5$ and $F1 < F3 < F2 < F4 < F5$ with 10 and 9 votes respectively. Among models representing interactional realism $I1 < I2 < I3 < I4 < I5$ was the most popular with 7 votes.

4 DISCUSSION

Main analyses revealed that the models significantly differ from each other. The Cronbach's alpha analyses showed that the internal consistency of the subscales is of very high rate. This leads to the conclusion that the items used in the subscales are redundant and in the future versions of the tool we should aim at increasing the variation among them. The test-retest analyses showed that the scores are stable over trials.

Qualitative analysis of ordering of the models follows the linear pattern. Among photographic realism types, two of the most popular orderings differ only in terms of place 3 and 4 which is difficult to understand, bearing in mind the difference in median ratings (fig. 3, supplementary materials). Nevertheless, in this type of realism we partially (except F1 and F2) achieved a consistent manipulation of level of details, numbers of triangles, resolution of textures, render pipelines and shaders, which resulted in almost linear pattern of median ratings (fig. 3, supplementary materials). Most frequent ordering followed the sequence marked out by the plot of median ratings and reflected the manipulation of the aforementioned variables. This shows that there might be a common factor which can be manipulated between the studies to achieve different levels of realism of appearance. Zell et al. [22] found that the material (i.e. combination of shaders, shader parameters and textures) of the character's face was the predominant cause of rating how appealing the character is and the face shape was the factor influencing realism ratings. Interestingly, our manipulation was focused on the material rather than shape which remained constant across conditions and still the differences in the realism ratings were achieved.

In case of realism of behavior, the two most frequent orderings differ only in terms of two of the first places. It seems that a static agent might be perceived as more realistic than a agent behaving in an unnatural way. Nevertheless, both in terms of median ratings (fig. 4, supplementary materials) and orderings the top 2 were the models maintaining eye contact with the participant. Bailenson and

associates [2] showed that agent displaying eye contact evoked significantly higher level of social presence (copresence) which is under the influence of realism [19]. Median ratings as well as ordering of agents from our study seem to overlap with that prediction in some way, because agents with the highest ratings and places in behavioral realism were those which maintained an eye contact. This is further supported by Borland, Peck and Slater's [3] study which underlines the importance of realistic eye movements on the quality of virtual experience. When it comes to the interactional realism the ordering follows the sequence based on median ratings (fig. 5, supplementary materials). Following Swinth and Blascovich [19] proposal that the more interactive character the better, we achieved a result in which the highest rating was granted to the character which directly addressed the participant.

This version of the tool can be considered preliminary and should be followed by updates with new features until the attainment of the desired, final instrument is achieved. This practice is in compliance with the current guidelines [13].

5 LIMITATIONS AND FUTURE DIRECTIONS

One of the key limitations is of our research is that the scale *per se* was created for the purpose of this research and was never psychometrically validated. To construct a complete tool to be used by wider audience we recommend thorough psychometric exploration. As indicated by Cronbach's alpha analyses, the structure of the scale at present state is unsatisfactory due to redundancy of the items in the subscales. In the future versions of the tool re-examination of the items should be conducted. Furthermore, the responses to the current questionnaire were presented to the participants only in arithmetic order and explained at the beginning of the online procedure, which may have increased the level of subjectivity of the responses. In future versions of the tool, each of the numbers might be matched with its respective description i.e. 1 - Highly Unrealistic, 2 - Very Unrealistic etc. and presented in that manner to the participants.

Furthermore, in this paper we focused specifically on the realism, but there are many other aspects of virtual agents that can be standardized. In the future, the tool might include other questionnaires e.g. those extracted by Filtrianie and associates [8].

We think that it would be beneficial to combine the tool with screening questions to include contextual and individual factors influencing responses in the questionnaire. Additionally it would be important to add the open-ended questions in the questionnaire to detect important features which might not be spotted with closed-ended questions. Moreover, we think that it would be interesting to combine the tool with methods such as eye-tracking or psychophysiology to correlate the questionnaire scores with the indirect measurements.

One of the major flaws of our study was small sample size which forced us to use non-parametric alternatives of the statistical tests. Although, we can observe a general linear trend in the data (fig. 3, 4, 5, supplementary materials) it needs to be statistically confirmed with e.g. contrast analyses under the ANOVA. Future versions of the tool should be tested on bigger samples with parametric tests usage. In this study, the sample size was due to the high competence level of the users asked for participation, however the online application can serve as a crowd-sourcing tool for gathering feedback from numerous users. The additional factors on the user side (display quality, size, etc.) were not controlled in this study.

Even though, the main analyses showed significant differences between the models it should be noted that the differences were relatively easy to spot. In the future, the tool might be tested on the stimuli differing to a lesser extent. Moreover, in this study we used three types of realism [19] which is not a dominating approach and was originally developed for 3D avatars and not agents. In the future, different theoretical models might be used for software development. When it comes to interactional realism, although the ratings of the I1-I5 models significantly differed which implicates a

sense of interactivity achieved, the character's did not present real interactions with the users as the lines were pre-recorded and played without the ability of the user to respond.

Even though we tried to achieve similar levels of brightness of the models, we did not control for that quantitatively. As shown by researchers [21] the exposure of the virtual character's face can significantly alter its perception (appealingness). Although the relationship between how realistic the agent is and how appealing it seems to be is complex [23], in the future this factor should be taken into the account.

One of the requirements of successful pretest is a measurement tool which will be widely used. We recommend further exploration in this topic and encourage the researchers to comment and add on to the work presented in this paper.

Although the tool was meant to be used in 3D character standardization in the current version it is a 2D web-based programme which does not take into account immersiveness of the medium, which can significantly influence perception of the virtual characters [15, 23]. In the future versions, it might be transferred to HMD goggles. WebXR API will be applied to provide immersive environment for viewing the models. We plan to develop some additional functionalities of model viewing, i.e. allowing the user to zoom and rotate the model freely - this way they can focus on the aspects they are most interested in and can use the zoom level most appropriate to rate a certain feature. In order to make the tool more universal, uploading own virtual objects or models from different databases (e.g. Rocketbox [9] or MakeHuman [1]) as well as uploading questionnaire items and setting up the randomization and displaying procedure will be implemented in further versions. We also plan to release the code to the scientific community.

6 CONCLUSIONS

The tool presented in this article is preliminary step towards better quality of VR research. Firstly, we wanted to bring the topic of pretesting to the discussion as we believe in its importance for conducting empirically grounded studies in Virtual Environments. Secondly, we presented a preliminary version of the tool which aimed at helping researchers perform the pretests on their own remotely i.e. even in a *pandemia* situation. Making pretesting easier and more accessible might result in studies becoming comparable and reproducible. The tool *per se* deals with complexity-efficiency trade-off, because at the same time combines state-of-the-art Unity technology and can be easy to implement. The tool can provide several measures which can be interpreted both quantitatively (ratings) and qualitatively (ordering). Although, the tool at the current form is not yet eligible for wider usage, we believe that in the future, valid versions will be developed which will bring about a major benefit to the VR research community.

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